ISyE 3770 HW4

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2024-09-29

Problem 3.8.12

Part A:

$$X \sim Pois(1.8)$$

$$P(X > 5) = 1 - P(X \leq 5) = 1 - \sum_{k=0}^{5} \frac{e^{-1.8} 1.8^k}{k!}$$

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1 - sum(exp(-1.8) * 1.8^(0:5) / factorial(0:5))
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## [1] 0.01037804
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Part B:

$$X \sim Pois(1.8 \times 7) = Pois(12.6)$$

$$P(X < 5) = \sum_{k=0}^{4} \frac{e^{-12.6} 12.6^k}{k!}$$

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sum(exp(-12.6) * 12.6^(0:4) / factorial(0:4))
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## [1] 0.004979028
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Part C:

$$X \sim Pois(1.8 \times d) = Pois(1.8d)$$

$$P(X \geq 1) = 1 - P(X < 1) = 1 - \frac{e^{-1.8d} (1.8d)^0}{0!} = 0.99$$

$$d = -\frac{ln(0.01)}{1.8}$$

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(-1) * log(0.01) / 1.8
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## [1] 2.558428
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Part D:

$$X \sim Pois(m \times 1) = Pois(m)$$

$$P(X > 5) = 1 - P(X \leq 5) = 1 - \sum_{k=0}^{5} \frac{e^{-m} m^k}{k!} = 0.1$$

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uniroot(function(m) 1 - sum(exp(-m) * m^(0:5) / factorial(0:5)) - 0.1, lower = 0, upper = 10)$root
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## [1] 3.151886
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Problem 4.1.2

Part A:

$$P(X < 2) = \int_1^2 \frac{2}{x^3} dx = \frac{3}{4}$$

Part B:

$$P(X > 5) = \int_5^\infty \frac{2}{x^3} dx = \frac{1}{25}$$

Part C:

$$P(4 < X < 8) = \int_4^8 \frac{2}{x^3} dx = \frac{3}{64}$$

Part D:

$$P(X < 4 \ or \ X > 8) = P(X < 4) + P(X > 8) = \int_8^\infty \frac{2}{x^3} dx + \int_1^4 \frac{2}{x^3} dx = \frac{1}{64} + \frac{15}{16} = \frac{61}{64}$$

Part E:

$$P(X < x) = \int_1^x \frac{2}{x^3} dx = 0.95$$

$$(-\frac{1}{x^2})\Big|_1^x = -\frac{1}{x^2} + 1 = 0.95$$
$$x = \sqrt{20}$$

Problem 4.4.2

Part A:

$$F(x) = \begin{cases} 0 & x < 0.95 \\ \frac{x-0.95}{1.05-0.95} = 10(x - 0.95) & 0.95 \leq x \leq 1.05 \\ 1 & x > 1.05 \end{cases}$$

Part B:

$$P(X > 1.02) = 1 - F(1.02) = 1 - 10(1.02 - 0.95) = 1 - 0.7 = 0.3$$

Part C:

$$P(X > x) = 0.9 \ or \ F(x) = 0.1$$
$$10(x - 0.95) = 0.1$$

$$x = 0.96$$

Part D:

$$\mu = \frac{a+b}{2} = \frac{0.95 + 1.05}{2} = 1.00$$
$$\sigma^2 = \frac{(b-a)^2}{12} = \frac{(1.05 - 0.95)^2}{12} = \frac{(0.1)^2}{12} = 0.000833$$

Problem 4.5.6

Part A:

$$Z = \frac{X - \mu}{\sigma} = \frac{240 - 260}{50} = -0.4$$
$$P(X > 240) = 1 - P(Z < -0.4) = 1 - 0.3446 = 0.6554$$

Part B:

For the first Quartile:

$$Z \approx -0.6745$$
$$X = \mu + Z(\sigma) = 260 + (-0.6745)(50) = 226.275 \ min$$

For the second Quartile:

$$Z \approx 0.6745$$
$$X = 260 + (0.6745)(50) = 293.7 \ min$$

Part C:

$$Z \approx -1.645$$
$$X = 260 + (-1.645)(50) = 177.75 \ min$$

Problem 4.5.12

Part A:

$$Z = \frac{X - \mu}{\sigma} = \frac{350 - 310}{45} = 0.8889$$

$$P(Z > 0.8889) = 1 - P(Z < 0.8889) = 0.1864$$

Part B:

$$Z \approx 2.326$$
$$X = 310 + 2.326(45) = 414.67 \ mill \ gal$$

Part C:

$$Z \approx -1.645$$
$$X = 310 + (-1.645)(45) = 235.975 \; mill \; gal$$

Part D:

$$Z = \frac{X - \mu}{45}$$
$$2.33 = \frac{350 + \mu}{45}$$
$$\mu = 245.2$$

Problem 4.7.5

Part A:

$$P(T \leq 30) = 1 - P(T > 30) = 1 - 1 + e^{-\frac{1}{15}(30)} = 0.1353$$

Part B:

$$P(T \leq 10) = 1 - e^{-\frac{1}{15}(10)} = 0.4866$$

Part C:

$$P(5 < T < 10) = P(T < 10) - P(T < 5) = 0.4866 - (1 - e^{-\frac{1}{15}(5)}) = 0.4866 - 0.2835 = 0.2031$$

Part D:

$$P(T \leq t) = 1 - e^{-\lambda t} = 0.90$$
$$1 - e^{-\frac{1}{15}t} = 0.90$$
$$t = 34.54 \; min$$

Problem 4.7.11

Part A:

$$\lambda t = \frac{1}{10}(30) = 3$$
$$N \sim Pois(3)$$
$$P(N > 3) = 1 - P(N \leq 3) = 1 - \sum_{k=0}^{3} P(N = k)$$

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1 - (exp(-3) + 3 * exp(-3) + (9 * exp(-3)) / 2 + (27 * exp(-3)) / 6)
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## [1] 0.3527681
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Part B:

As previously calculated:

$$P(N = 0) = e^{-3} = 0.0498$$

Part C:

$$e^{-\frac{1}{10}(x)(60)} = 0.01$$
$$x = \frac{4.605(10)}{6} = 7.68\ hours$$

Part D:

$$P(no\ calls\ in\ 120m) = e^{-\frac{1}{10}(120)} = 0.00000614$$

Question 2

Part A: CDF

$$F(x) = \int_1^x \frac{2}{t^3}dt = [-\frac{1}{t^2}]\Big|_1^x = -\frac{1}{x^2} + 1,\ for\ x > 1$$

$$F(x) = \begin{cases} 0 & x \leq 1 \\ 1 - \frac{1}{x^2} & x > 1 \end{cases}$$

Part B: Mean

$$\mu = \int_1^\infty x(\frac{2}{x^3})dx = -2[\frac{1}{x}]\Big|_1^\infty = 2$$

Part C: Variance

$$Var(X) = E[X^2] - (E[X])^2$$

$$E[X^2] = \int_1^\infty x^2(\frac{2}{x^3})dx = \int_1^\infty \frac{2}{x}dx = [2ln(x)]\Big|_1^\infty = \infty$$

Therefore, since $E[X^2] = \infty$, the variance is infinite.